

Clinical Natural Language Technology for Health Care

From Rule-Based Pipelines to Reasoning Models and What It Means for Payment Integrity

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1. Clinical NLP and How It Evolved

Clinical natural language processing (NLP) is the set of methods that turn unstructured medical text, such as progress notes, discharge summaries, and pathology reports, into structured data a computer can act on. Most of a patient's story lives in free text, not in coded fields, which is the whole reason the field exists.

The first working systems were rule-based. MetaMap, released by the National Library of Medicine, mapped phrases in clinical text to concepts in the UMLS Metathesaurus using hand-built lexical rules (Aronson & Lang, 2010). cTAKES, built at Mayo Clinic, chained together tokenizers, part-of-speech taggers, and dictionary lookups to pull entities like diagnoses and medications out of records (Savova et al., 2010). These tools were transparent and auditable. They were also brittle. Change the phrasing or add an abbreviation and recall fell off.

The transformer architecture changed the economics of that problem. BERT learned language representations from context instead of fixed rules (Devlin et al., 2019), and within a year domain-specific versions appeared. BioBERT, pretrained on PubMed abstracts, beat general models on biomedical entity recognition (Lee et al., 2020). ClinicalBERT trained on actual hospital notes from MIMIC-III (Alsentzer et al., 2019). The payoff was generalization. A model that has read millions of sentences knows that “MI,” “myocardial infarction,” and “heart attack” point at the same thing without anyone writing that rule by hand.

Large language models pushed the line again. GPT-4 cleared the USMLE passing threshold by more than 20 points with no medical fine-tuning (Nori et al., 2023), and Google's Med-PaLM reached expert-level performance on medical question answering (Singhal et al., 2023). The newest models are multimodal. Med-Gemini reads images and text together and outperformed GPT-4V on multimodal medical benchmarks by a wide margin (Saab et al., 2024). That matters for billing work, because optical character recognition (OCR) and computer vision are the bridge from paper to text. A scanned chart or a handwritten order is just pixels until a vision model reads it. Once it does, every downstream NLP method becomes available.

2. Where the Field Is Heading

A few trends are worth watching. Ambient documentation is the clearest near-term shift: at Kaiser Permanente an ambient AI scribe that listens to the visit and drafts the note was used in more than 300,000 patient encounters in its first ten weeks, and physicians reported less after-hours charting (Tierney et al., 2024). The second is multimodal reading of messy inputs, including handwriting, which until recently broke OCR pipelines. The third, and the one that matters most for payment integrity, is the move from extraction to reasoning. Older systems answered, “what entities are in this note.” Current models can be asked “does the documentation support this level of service,” which is a judgment rather than a lookup.

3. Opportunities and Threats for Cotiviti

That judgment is exactly where Cotiviti operates. The opportunity is direct. A model that reads the clinical chart and the submitted claim, then flags where they disagree, scales review well past what human auditors can cover. Payment integrity today samples claims because manual review is expensive; a reviewer-assist model lets you screen far more of them.

The threats are real, and I would not wave them away. LLMs hallucinate. They generate fluent text that is not grounded in the source, and a survey of the problem found it is intrinsic to how these models are trained, not a bug that patches away (Ji et al., 2023). A model that invents a diagnosis it never saw is worse than useless in an audit. PHI is the second problem: protected health information cannot leave a covered entity's control without a business associate agreement, so piping charts to a third-party API can violate HIPAA (U.S. Department of Health and Human Services, 2013). Third is audit defensibility. A denial must survive appeal, and "the model said so" will not survive anything. You need the specific sentence the decision rests on. As regulators sharpen their attention on AI in coverage decisions, an unexplainable model becomes a liability rather than an asset.

4. Recommendation

So here is my position. Cotiviti should pilot a reviewer-assist mismatch detector, not an autonomous denial engine, and it should run on models covered by BAA or hosted on-premises. Human-in-the-loop is not a hedge here. It is the design choice that makes the threats above manageable: the model proposes, a trained auditor disposes, and every flag carries its evidence.

To test whether the core idea holds, I built a working demo called ClaimGuard Lite. It takes a chart image and a claim image, runs OCR with a vision model, and extracts structured facts (diagnosis, medications, procedures, severity, and billed amount), attaching the exact source sentence to each one. It then compares the chart against the bill and returns a risk score and a recommended action: auto-approve, human review, or escalate to an auditor. In one test I fed it a chart documenting a same-day upper respiratory infection treated with ibuprofen, against a bill coding pneumonia with high complexity, a chest x-ray, three medications, and a \$12,400 charge. The tool caught five mismatches and escalated it. I built it two ways on purpose: a rule-based keyword matcher standing in for the old approach, and Claude Haiku 4.5 for the LLM approach, with the system falling back to rules when no API key is present.

I will be honest about what it is not. The keyword engine is a baseline, not a real NLP system. It does no ICD-10 or CPT code validation, it runs on text demos rather than live claims, and it is nowhere near production. What it does show is the shape of the right product: an explainable, human-supervised tool that reads both documents and points an auditor straight at the disagreement. That is the version of this technology Cotiviti should build, and the version I would want to defend a denial on appeal.

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